

Отчет по практической работе №2

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Группа: БСБО-16-23

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1. Информация о кластере

1.1 Статус Minikube

```
Командная строка
Microsoft Windows [Version 10.0.26200.8037]
(c) Корпорация Майкрософт (Microsoft Corporation). Все права защищены.

C:\Users\Dani>curl.exe -LO "https://dl.k8s.io/release/v1.28.0/bin/windows/amd64/kubectl.exe"
 % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
             Dload  Upload   Total      Spent    Left     Speed
100 48.23M 100 48.23M    0      0 1.52M      0    0:31    0:31    0:31   1.84M

MINGW64:/c/Projects/devops/lab2
Dani@DaniPC MINGW64 /c/Projects/devops/lab2 (main)
$ git push --set-upstream origin main
bash: '$\302\203git': command not found

Dani@DaniPC MINGW64 /c/Projects/devops/lab2 (main)
$ git push -u origin main
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 20 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 1.17 KiB | 1.17 MiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/DaniF77/containers-lab-2
   8c7081b..44d911e  main -> main
branch 'main' set up to track 'origin/main'.

Dani@DaniPC MINGW64 /c/Projects/devops/lab2 (main)
$ kubectl version --client
Client Version: v1.34.1
Kustomize Version: v5.7.1

Dani@DaniPC MINGW64 /c/Projects/devops/lab2 (main)
$
```

1.2 Узлы кластера

```
Dani@DaniPC MINGW64 /c/Projects/devops/lab2/k8s (lab2/kubernetes-basics)
$ kubectl get nodes -o wide
NAME          STATUS    ROLES          AGE      VERSION    INTERNAL-IP    EXTERNAL-IP
OS-IMAGE      KERNEL-VERSION CONTAINER
minikube      Ready     control-plane  7m31s    v1.35.1    192.168.49.2   <none>
Debian GNU/Linux 12 (bookworm) 6.6.87.2-microsoft-standard-WSL2 docker://29.2.1
```

2. Созданные ресурсы

2.1 Pods

```
Dani@DaniPC MINGW64 /c/Projects/devops/lab2/k8s (lab2/kubernetes-basics)
$ kubectl get pods -o wide
NAME          READY   STATUS    RESTARTS   AGE   IP            NODE       NOMINATED
NODE   READINESS GATES
nginx-pod     1/1     Running   0           64s   10.244.0.6    minikube   <none>
<none>
```

```
Dani@DaniPC MINGW64 /c/Projects/devops/lab2/k8s (lab2/kubernetes-basics)
$ |
```

2.2 Deployments

```
Dani@DaniPC MINGW64 /c/Projects/devops/lab2/k8s (lab2/kubernetes-basics)
$ kubectl get deployments
No resources found in default namespace.
```

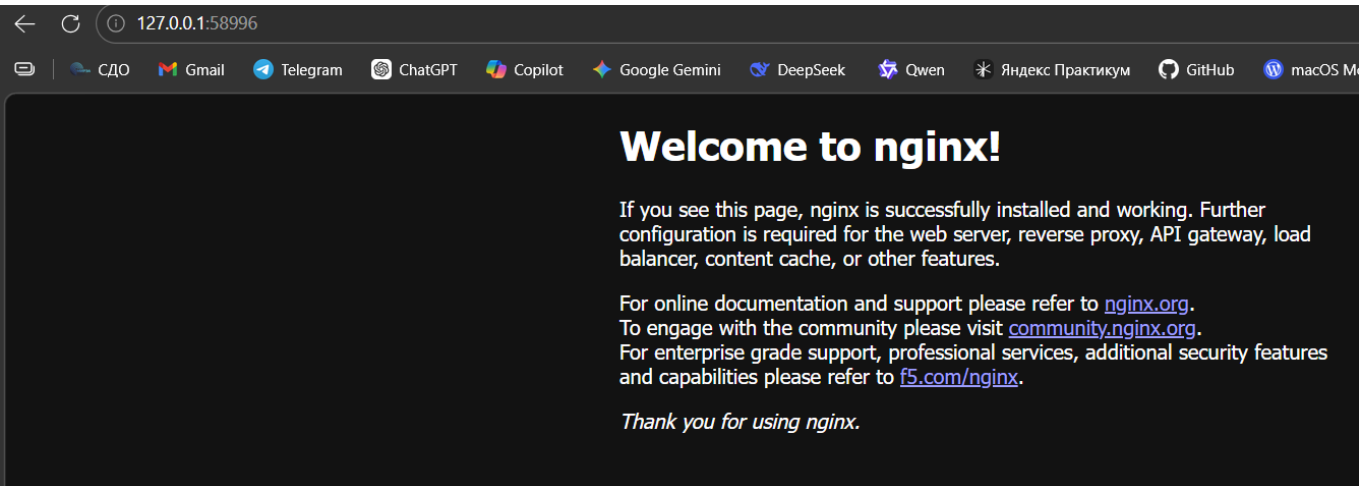
так как мы создали не deployment, а Pod напрямую

2.3 Services

[вставьте вывод kubectl get services]

```
Dani@DaniPC MINGW64 /c/Projects/devops/lab2/k8s (lab2/kubernetes-basics)
$ kubectl get services
NAME            TYPE        CLUSTER-IP   EXTERNAL-IP   PORT(S)          AGE
kubernetes      ClusterIP   10.96.0.1    <none>        443/TCP          25m
nginx-deployment NodePort    10.97.24.210 <none>        80:30398/TCP     39s
```

```
Dani@DaniPC MINGW64 /c/Projects/devops/lab2/k8s (lab2/kubernetes-basics)
```

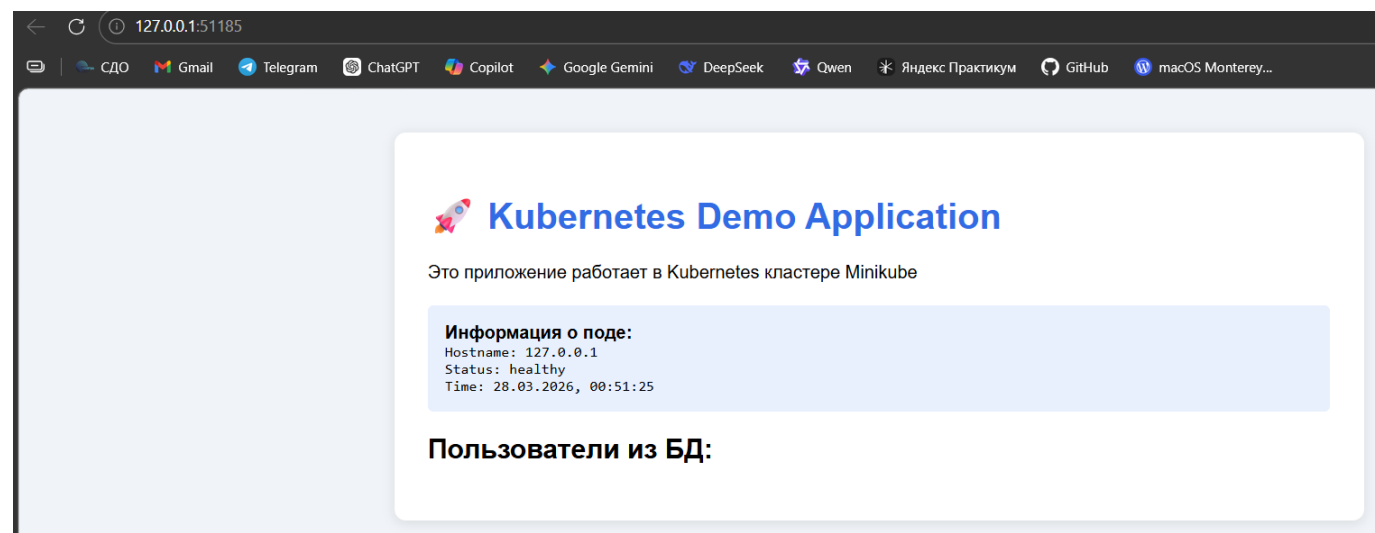


2.4

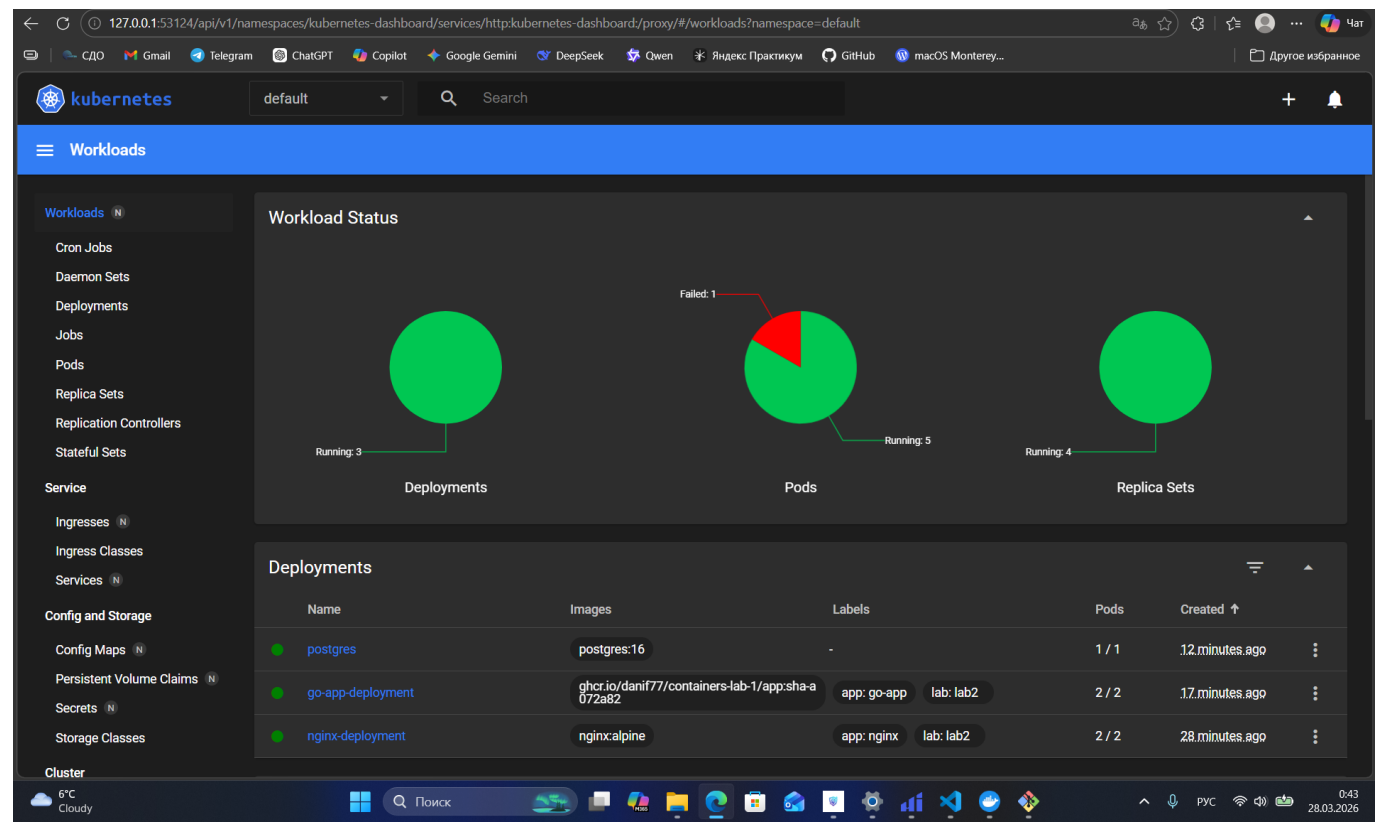
```
Dani@DaniPC MINGW64 /c/Projects/devops/lab2/k8s (lab2/kubernetes-basics)
$ kubectl logs -l app=go-app
2026/03/27 21:29:03 Error creating table:dial tcp: lookup postgres-service on 10
.96.0.10:53: server misbehaving
2026/03/27 21:25:50 Error creating table:dial tcp [::1]:5432: connect: connectio
n refused
2026/03/27 21:29:06 Error creating table:dial tcp: lookup postgres-service on 10
.96.0.10:53: server misbehaving
```

3. Скриншоты работы приложения

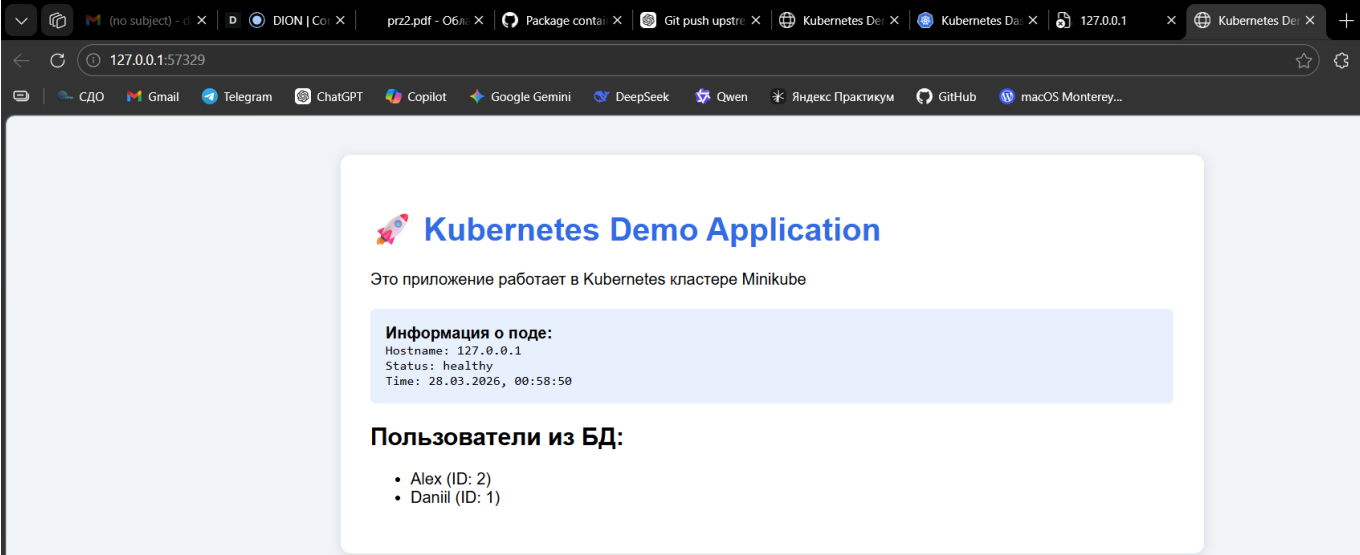
3.1 Главная страница



3.2 Дашборд Kubernetes



3.3 Результат GET /api/users



вывод пользователей

4. Эксперименты с масштабированием

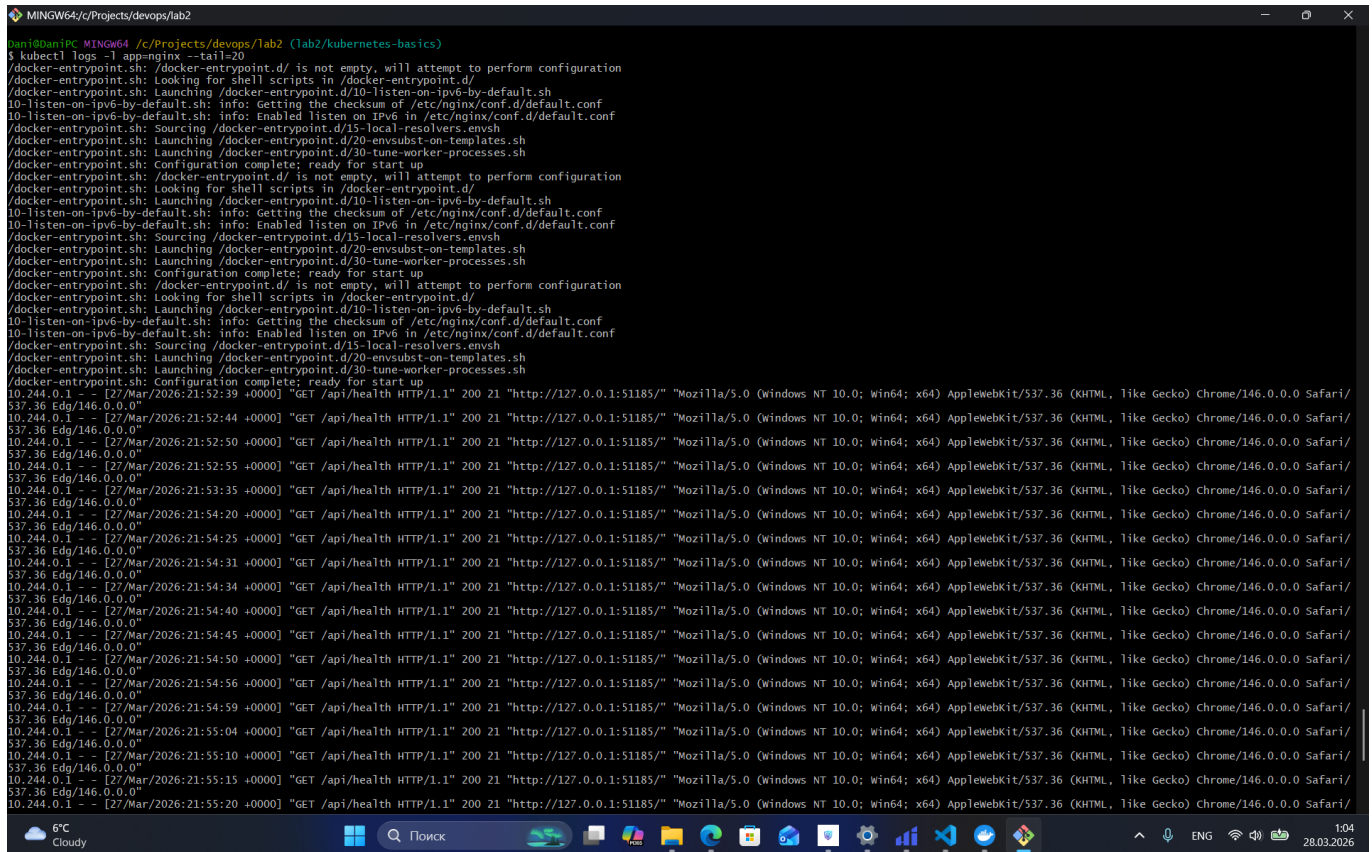
4.1 Масштабирование до 5 реплик

```
Dani@DaniPC MINGW64 /c/Projects/devops/lab2 (lab2/kubernetes-basics)
$ kubectl scale deployment nginx-deployment --replicas=5
deployment.apps/nginx-deployment scaled

Dani@DaniPC MINGW64 /c/Projects/devops/lab2 (lab2/kubernetes-basics)
$ kubectl get pods
NAME                                READY   STATUS    RESTARTS
AGE
debug                               1/1     Running   0
7m49s
go-app                              0/1     CrashLoopBackOff   15 (111s ago)
53m
go-app-deployment-78876cd4d4-lw7bd  1/1     Running   8 (26m ago)
37m
go-app-deployment-78876cd4d4-w47vs  1/1     Running   8 (26m ago)
37m
nginx-deployment-795ccb77f-5nk8t    1/1     Running   0
26s
nginx-deployment-795ccb77f-cpg4n    1/1     Running   0
26s
nginx-deployment-795ccb77f-h2zc2    1/1     Running   0
26s
nginx-deployment-795ccb77f-ng86t    1/1     Running   0
13m
nginx-deployment-795ccb77f-qxqw4    1/1     Running   0
13m
postgres-78877d5b4-l8jrv           1/1     Running   0
32m

Dani@DaniPC MINGW64 /c/Projects/devops/lab2 (lab2/kubernetes-basics)
$ |
```

4.2 Проверка распределения нагрузки



```
banid@DaniPC MINGW64 /c:/Projects/devops/lab2 (lab2/kubernetes-basics)
$ kubectl logs -l app=nginx --tail=20
/docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform configuration
/docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
/docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh
10-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/nginx/conf.d/default.conf
10-listen-on-ipv6-by-default.sh: info: Enabled listen on IPv6 in /etc/nginx/conf.d/default.conf
/docker-entrypoint.sh: Sourcing /docker-entrypoint.d/15-local-resolvers.envsh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-processes.sh
/docker-entrypoint.sh: Configuration complete; ready for start up
/docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform configuration
/docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
/docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh
10-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/nginx/conf.d/default.conf
10-listen-on-ipv6-by-default.sh: info: Enabled listen on IPv6 in /etc/nginx/conf.d/default.conf
/docker-entrypoint.sh: Sourcing /docker-entrypoint.d/15-local-resolvers.envsh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-processes.sh
/docker-entrypoint.sh: Configuration complete; ready for start up
10.244.0.1 - [27/Mar/2026:21:52:39 +0000] "GET /api/health HTTP/1.1" 200 21 "http://127.0.0.1:51185/" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/146.0.0.0 Safari/537.36 Edg/146.0.0.0"
10.244.0.1 - [27/Mar/2026:21:52:44 +0000] "GET /api/health HTTP/1.1" 200 21 "http://127.0.0.1:51185/" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/146.0.0.0 Safari/537.36 Edg/146.0.0.0"
10.244.0.1 - [27/Mar/2026:21:52:50 +0000] "GET /api/health HTTP/1.1" 200 21 "http://127.0.0.1:51185/" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/146.0.0.0 Safari/537.36 Edg/146.0.0.0"
10.244.0.1 - [27/Mar/2026:21:52:55 +0000] "GET /api/health HTTP/1.1" 200 21 "http://127.0.0.1:51185/" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/146.0.0.0 Safari/537.36 Edg/146.0.0.0"
10.244.0.1 - [27/Mar/2026:21:53:05 +0000] "GET /api/health HTTP/1.1" 200 21 "http://127.0.0.1:51185/" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/146.0.0.0 Safari/537.36 Edg/146.0.0.0"
10.244.0.1 - [27/Mar/2026:21:54:20 +0000] "GET /api/health HTTP/1.1" 200 21 "http://127.0.0.1:51185/" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/146.0.0.0 Safari/537.36 Edg/146.0.0.0"
10.244.0.1 - [27/Mar/2026:21:54:25 +0000] "GET /api/health HTTP/1.1" 200 21 "http://127.0.0.1:51185/" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/146.0.0.0 Safari/537.36 Edg/146.0.0.0"
10.244.0.1 - [27/Mar/2026:21:54:31 +0000] "GET /api/health HTTP/1.1" 200 21 "http://127.0.0.1:51185/" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/146.0.0.0 Safari/537.36 Edg/146.0.0.0"
10.244.0.1 - [27/Mar/2026:21:54:40 +0000] "GET /api/health HTTP/1.1" 200 21 "http://127.0.0.1:51185/" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/146.0.0.0 Safari/537.36 Edg/146.0.0.0"
10.244.0.1 - [27/Mar/2026:21:54:45 +0000] "GET /api/health HTTP/1.1" 200 21 "http://127.0.0.1:51185/" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/146.0.0.0 Safari/537.36 Edg/146.0.0.0"
10.244.0.1 - [27/Mar/2026:21:54:50 +0000] "GET /api/health HTTP/1.1" 200 21 "http://127.0.0.1:51185/" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/146.0.0.0 Safari/537.36 Edg/146.0.0.0"
10.244.0.1 - [27/Mar/2026:21:54:56 +0000] "GET /api/health HTTP/1.1" 200 21 "http://127.0.0.1:51185/" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/146.0.0.0 Safari/537.36 Edg/146.0.0.0"
10.244.0.1 - [27/Mar/2026:21:54:59 +0000] "GET /api/health HTTP/1.1" 200 21 "http://127.0.0.1:51185/" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/146.0.0.0 Safari/537.36 Edg/146.0.0.0"
10.244.0.1 - [27/Mar/2026:21:55:04 +0000] "GET /api/health HTTP/1.1" 200 21 "http://127.0.0.1:51185/" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/146.0.0.0 Safari/537.36 Edg/146.0.0.0"
10.244.0.1 - [27/Mar/2026:21:55:10 +0000] "GET /api/health HTTP/1.1" 200 21 "http://127.0.0.1:51185/" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/146.0.0.0 Safari/537.36 Edg/146.0.0.0"
10.244.0.1 - [27/Mar/2026:21:55:15 +0000] "GET /api/health HTTP/1.1" 200 21 "http://127.0.0.1:51185/" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/146.0.0.0 Safari/537.36 Edg/146.0.0.0"
10.244.0.1 - [27/Mar/2026:21:55:20 +0000] "GET /api/health HTTP/1.1" 200 21 "http://127.0.0.1:51185/" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/146.0.0.0 Safari/537.36 Edg/146.0.0.0"
```

5. GitHub Actions

5.1 Успешная валидация манифестов



6. Ответы на контрольные вопросы

1. В чем разница между Pod и Deployment? Pod — это один (или несколько) контейнеров. Deployment — управляет Pod'ами: создаёт, обновляет и следит, чтобы нужное число работало.
2. Для чего нужен Service типа ClusterIP? Даёт внутренний IP в кластере для доступа к Pod'ам между собой (внутренний балансировщик, снаружи недоступен).
3. Как ReplicaSet обеспечивает самовосстановление? ReplicaSet следит за количеством Pod'ов. Если Pod падает/удаляется — он создаёт новый вместо него.
4. Что произойдет с приложением, если удалить под PostgreSQL? ReplicaSet/Deployment поднимет новый Pod автоматически, данные сохранятся только если есть persistent volume, иначе могут потеряться.

7. Выводы

В ходе выполнения практической работы №2 были изучены базовые принципы оркестрации контейнеризованных приложений. Локальный кластер Minikube развернут согласно выданным методическим указаниям, требуемые YAML-манифесты созданы и применены без критических сбоев. Программа лабораторной работы выполнена в полном объеме, цели достигнуты.